

## INTRODUCTION

We want to look at internals of UNIX:

File Management

Memory Management.

Process Management

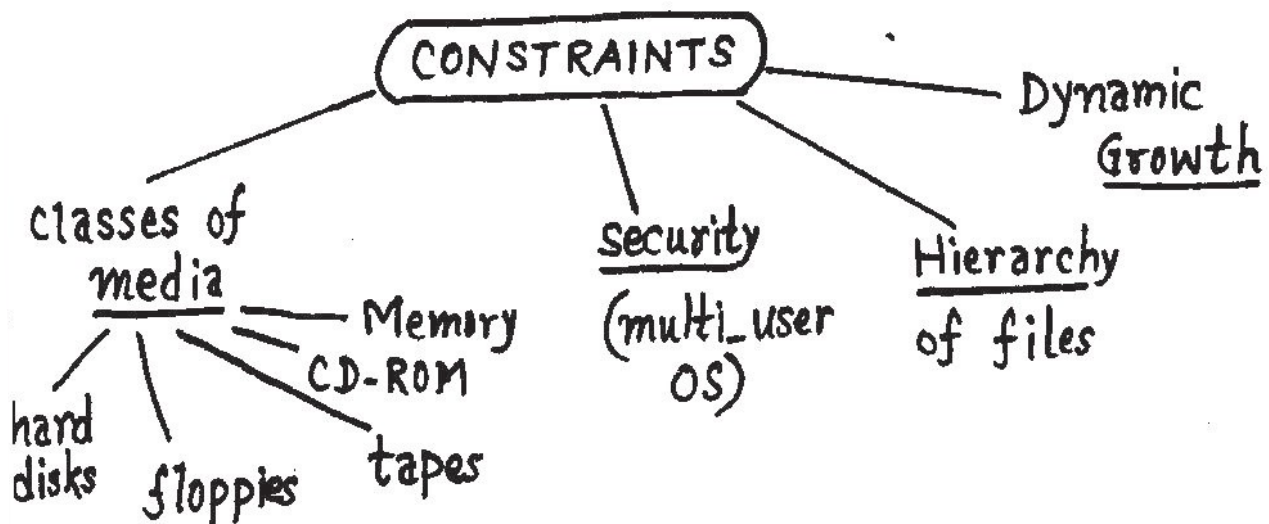
Inter Process Communication, etc.

## FILE MANAGEMENT

ACCESS TIME  
is the key

File management : the way OS organizes files

NOTE: UNIX file management usually  
much faster than DOS



Need to look at FMS implementation details.

Let us raise relevant questions today.

### MEDIUM : FLOPPY

Small capacities : 1.2 MB (5 1/4" drive)  
1.44 MB (3 1/2" drive)

### MEDIUM: HARD DISKS

#### Technologies

Winchester (Hard Fixed Disk)  
Removeable Hard Disk (less popular)  
: useful to store versions of OS  
: useful for demos and Road Shows

#### Capacities

10 MB

⋮

Giga bytes

Explosive growth  
FMS should scale properly

#### Interface

Connector  
to  
Hard Disk

ST 506

SMD

SASI

SCSI

IDE

### MEDIUM : CD-ROM WORM.

cheap to manufacture ; Fairly slow  
large capacities : 650 MB

### CD-ROM : USAGE

#### (a) Software Distribution

Ex: Early versions of UNIX : 10 - 20 MB  
(floppies used)

Latest versions of UNIX : 100 - 140 MB  
(Distribution on CD : popular)

#### (b) Unchanging / Slowly changing informations

Ex: Databases in Metros  
Music

### MEDIUM : TAPES

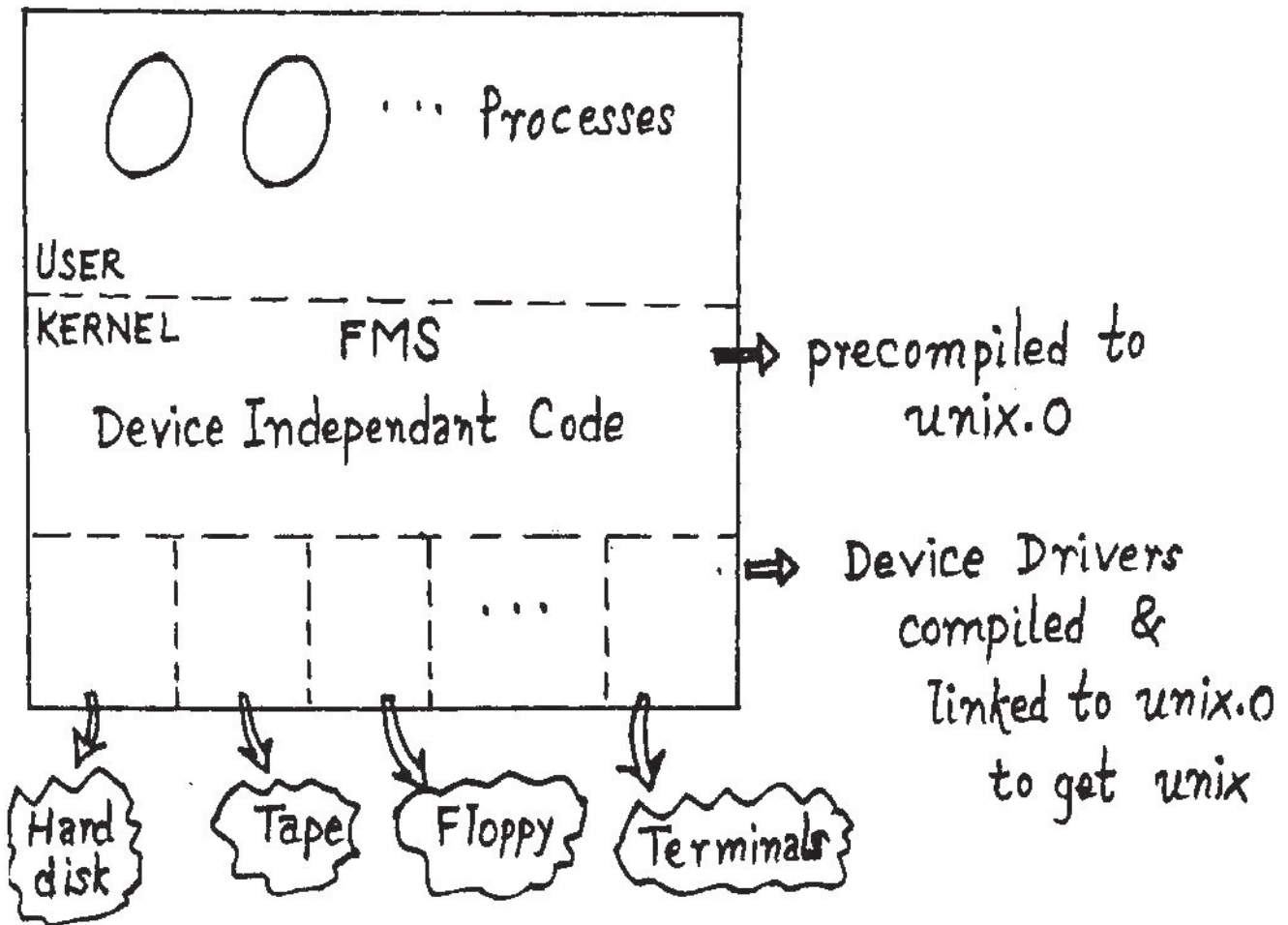
sequential access  
useful for taking backups.

### HETEROGENEOUS DEVICES

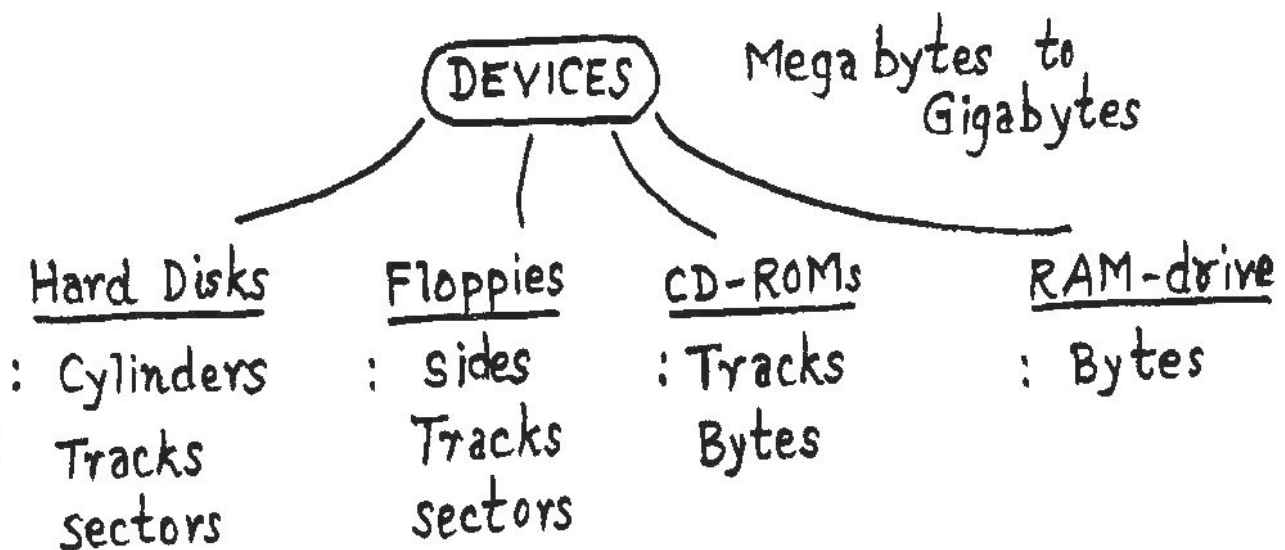
CD-ROM : not even manufactured  
when FMS was designed

OS : How it handles different devices ?

## Heterogeneous Devices



Each class of devices needs a device driver.





## (Heterogeneous Devices)

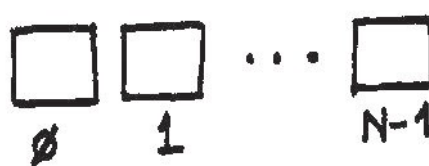
Device Driver : gives logical picture of the device.

core memory : can be used as RAM - drive  
Superfast ; not used under UNIX

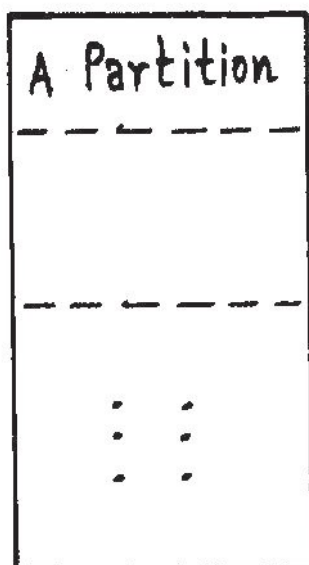
RAM - drive : Extension of Memory as drive

swap device : Extension of drive as memory

Uniform Picture of a device to FMS :

 : A stream of Bytes

(PARTITIONING) of hard disks  
because of large capacities.

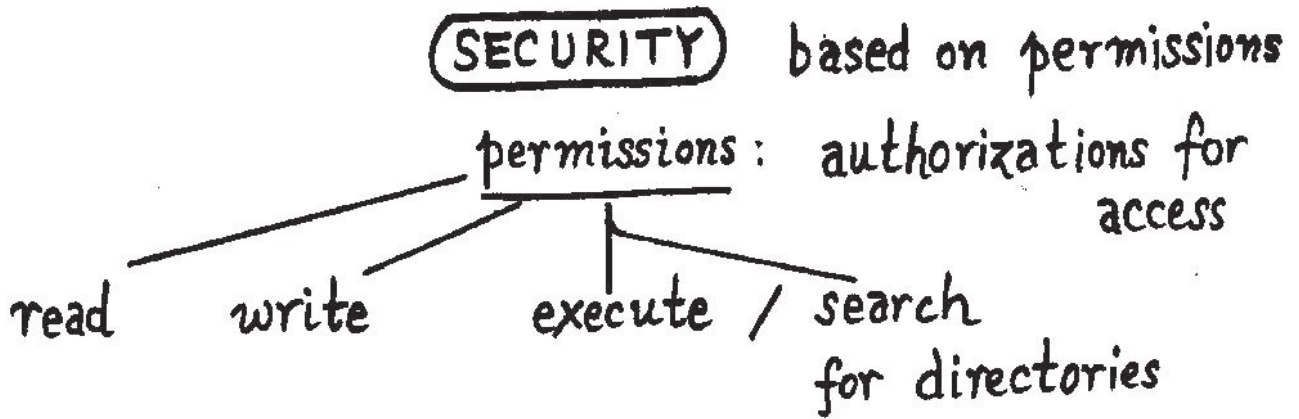


Each partition : a byte stream

FMS : stores a file system  
in a partition

: can operate on a  
forest of trees

Hierarchy : Directories treated  
as files.

Problem:

Hundreds of users  
Thousands of files }  $\Rightarrow$  millions of permissions

security checks : must not slow down system

Analogy: Me, My Family, Others

Each File: Owner, User Group, Other Users

Groups: (A) Different applications like payroll, finance, etc.

(B) Different Project files

We need: TWX rwx rwx : 9 permission bits per file

**IMPLEMENTATION**

How FMS implements a file system on a partition of a hard disk

### A PARTITION

Looks like a stream of bytes to FMS

Analogy: Book has several chapters  
File System has several files;  
but has dynamic growth.

Security: cannot be handled by considering  
partition as a byte stream

hence: 0 1 ... N-1 a stream of  
blocks

Floppy / Hard Disk : do I/O one sector at a time

Logical block size : multiple/submultiple  
of sector size

A chapter has several pages  
starts on a fresh page

A File has several blocks  
starts on a fresh block

### BLOCK SIZE

characterizes  
file system

Ex: S51K , S52K, S54K : file system names

block size

like page size for a block

too large

Each chapter starts on  
a fresh page

Wastage:  $\frac{1}{2}$  block per file

too small

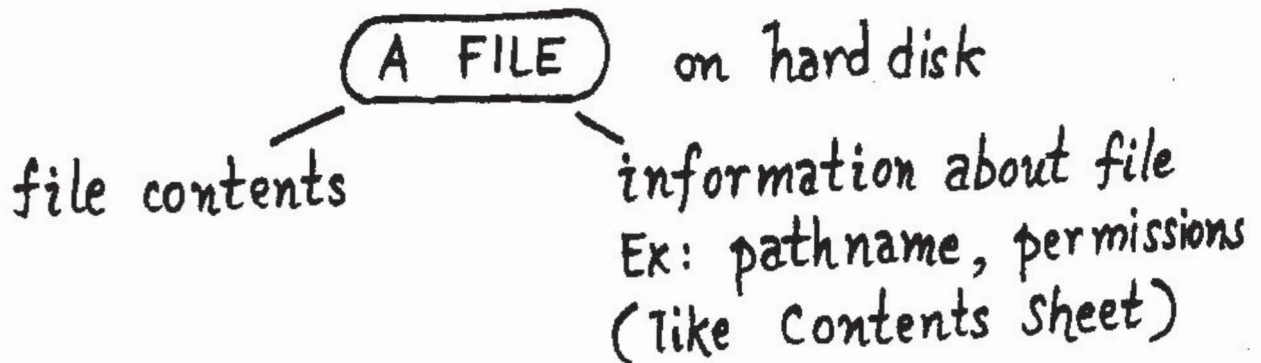
(a) Binding problem

(b) Access time slow

Engineering Decision

select block size that is as large as possible  
but with acceptable wastage.

=> has been increasing with  
increases in hard disk capacities

A Contents Sheet

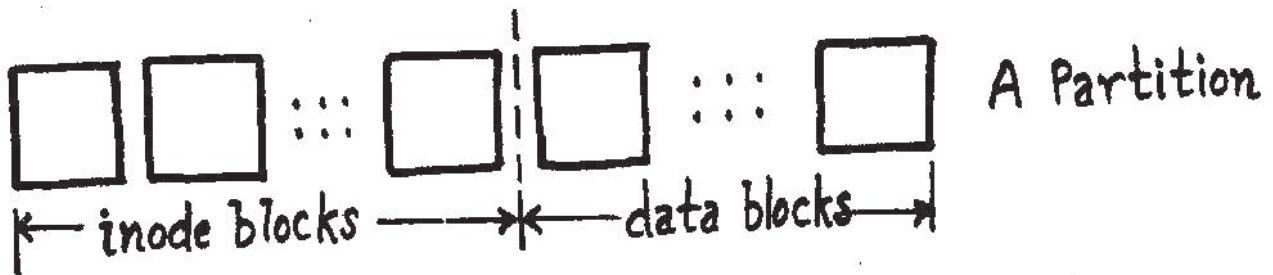
| <u>Sl. No.</u> | <u>Chapter</u> | <u>Page #</u> |
|----------------|----------------|---------------|
| 1.             | Introduction   | 1             |
| 2.             | Overview       | 5             |
| 3.             | Details ..     | 27            |

ONLY 1st page  
specified



a file

File System needs to contain the entire  
block number list for a file



Each File: Needs one inode block

Question: How to identify contents sheets in a book?

ANS: Page numbers use Roman literals

FMS: The split cannot be altered once fixed.

Usually, 25 % of partition reserved  
for inode blocks

EDP System : more data blocks

School System: more inode blocks

inode block number : FMS uses 2 bytes  
Max. 65,536 files in partition

data block number : FMS uses 3 bytes  
Max: 16 M blocks

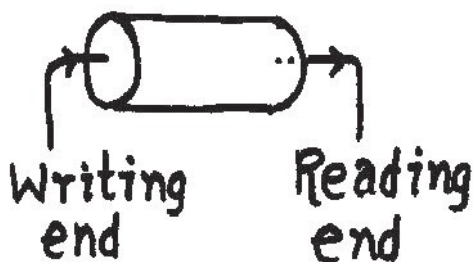
## INODE INFORMATION

Each file has one inode.

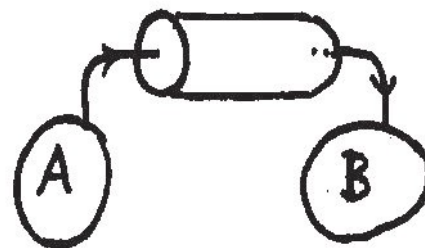
File Type FMS treats following as files:

- (a) ordinary files      Ex: sh, our files
- (b) directories
- (c) FIFO's      are named pipes

A pipe:



Simple IPC



(d) devices

Block devices : I/O in fixed size blocks

Ex: Hard disk, floppy

character devices : I/O is byte-oriented

Ex: Terminals, printers

Streams devices : I/O is message-oriented

Used for IPC

& networking

## inode informations

**Owner** of the file : Name not used

uid : user identification number

unsigned short (Max: 65,536 users)

Password file : contains other user details

## Permissions

Usual:      rwX    rwX    rwX    : 9 bits  
                       <sub>3</sub>            <sub>4</sub>                    }  
                       for owner    group            others

### Additional

(a) Sticky bit : Ex : vi, csh.  
                                          for frequently used executables

Copy of data blocks kept in swap device

(b) Set User-ID bit on Exec.

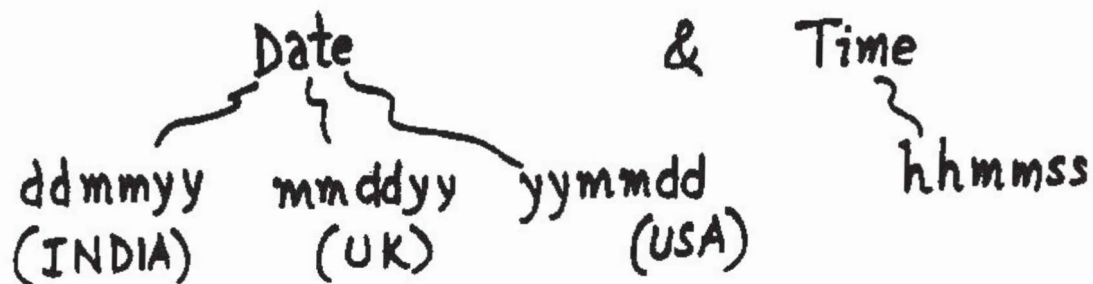
(c) Set Group-ID bit on Exec.

12 permission bits stored in inode of a file.

## inode informations

## Time Stamps

A time stamp is



Ex: 01.02.03 can mean:

Feb 1, 2003 (INDIA)

Feb 3, 2001 (USA)

Jan 2, 2003 (UK)

This human readable format difficult for comparisons, interest calculations, etc.

UNIX uses machine readable format: (unsigned long)

No. of Seconds since Jan 1, 1980

Read ctime(3).

## Time Stamps in inode

ctime

: creation time stamp  
of historic interest

mtime

modification  
time stamp

atime

access time stamp  
useful for Security  
& for cleanups